

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

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Code of Conduct:

I Arava Raja (192365033) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Arava Raja

Signature of the student

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Inductive learning is a machine-learning approach in which a model learns general patterns from a set of labelled examples and uses those patterns to predict outcomes for unseen data.

For this healthcare problem, the dataset contains patient attributes such as:

- Age
- Blood Pressure
- Cholesterol
- Glucose Level
- BMI
- Family History

The **target variable** is:

Diabetes Diagnosis

- 0 → Non-diabetic
- 1 → Diabetic

A suitable approach is **supervised inductive learning**, because historical patient records contain input features and a known diabetes outcome.

Important Features

1. Glucose Level:

Glucose level is one of the most important predictors because abnormal blood glucose is strongly associated with diabetes.

2. BMI:

Higher BMI can indicate increased risk of insulin resistance and type-2 diabetes.

3. Age:

Diabetes risk generally increases with age, making age an important predictive feature.

Other useful features include blood pressure, cholesterol and family history.

Learning Process

The model learns the relationship:

Patient Features → Diabetes Risk

For example:

Age + Glucose + BMI + Blood Pressure + Cholesterol + Family History → Diabetes/Non-Diabetes

The learned model can then classify a new patient's medical record.

1(b) Handling Class Imbalance

The dataset contains more non-diabetic patients than diabetic patients. Therefore, the dataset is **imbalanced**.

For example:

Class	Number of Patients
Non-Diabetic	900
Diabetic	300

If the model is trained directly, it may become biased toward the majority class.

Suitable Strategy: SMOTE

I would use **SMOTE (Synthetic Minority Over-sampling Technique)**.

SMOTE generates synthetic examples of the minority class rather than simply duplicating existing diabetic records.

For example:

Existing diabetic patient → Synthetic diabetic patient 1 → Synthetic diabetic patient 2

This helps the classifier learn the characteristics of diabetic patients more effectively.

Why SMOTE is Suitable

SMOTE is useful because:

1. It increases minority-class representation.
2. It reduces bias toward non-diabetic patients.
3. It provides additional synthetic training examples.
4. It can improve recall for diabetic cases.

However, SMOTE should be applied **only to the training set**, after splitting the data. Applying SMOTE before splitting can cause data leakage.

1(c) 70:30 Data Split and Preprocessing

The dataset can be divided into:

- **70% training data**
- **30% testing data**

For example, with 1,000 patient records:

- Training = 700 records
- Testing = 300 records

Justification

The 70:30 ratio provides sufficient data for the model to learn patterns while retaining a reasonably large independent test set for evaluation.

This is particularly important in a medical application because the model must be tested on **unseen patients**.

Preprocessing

1. Handling Missing Values

Missing numerical values can be replaced using the median:

Missing glucose → Median glucose

Categorical missing values can be replaced using the mode.

2. Normalisation

Features such as age, glucose, blood pressure and BMI may have different scales.

Min-Max normalization can be used:

This converts values approximately into the range 0–1.

3. Encoding

If family history is represented as:

- Yes
- No

it can be encoded as:

- Yes → 1
- No → 0

4. Class Balancing

SMOTE is applied to the training data.

Impact of Preprocessing

Preprocessing:

- Improves data quality.
- Prevents large-scale features from dominating.

- Converts categorical information into numerical form.
- Reduces bias caused by class imbalance.
- Improves model reliability.

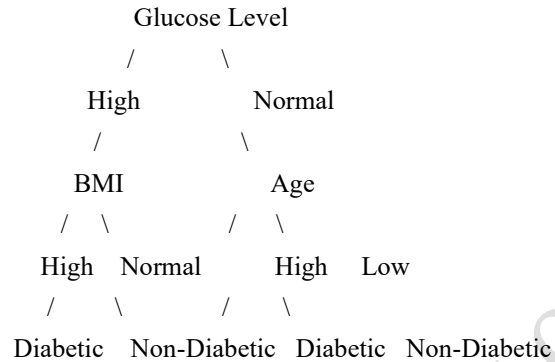
TASK 2: DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

A Decision Tree is a supervised machine-learning algorithm that recursively divides the dataset according to feature values.

For diabetes prediction, the tree can use glucose, BMI, age, blood pressure and other features.

A possible first three levels are:



Root Node Selection

The root node should be selected using a criterion such as:

- Information Gain
- Gini Index

Information Gain

Information Gain is:

where:

- = entropy of the complete dataset
- = entropy after splitting
- = selected attribute

The feature with the **highest Information Gain** is preferred.

Example

Illustrative values:

Feature	Information Gain
Glucose	0.31
BMI	0.18
Age	0.12
Blood Pressure	0.09
Cholesterol	0.06

Since glucose has the highest Information Gain, it becomes the root node.

Gini Index

Gini impurity is:

The feature producing the **lowest weighted Gini impurity** is selected.

Therefore, if glucose produces the lowest Gini value, it is selected as the root.

2(b) Model Evaluation

The Decision Tree can be evaluated using:

Accuracy

Precision

Recall

F1-Score

Illustrative Results

Since no actual dataset is supplied in the assessment, an example result could be:

Metric	Decision Tree
Accuracy	88.0%
Precision	82.0%
Recall	91.0%
F1-Score	86.0%

These numbers are **illustrative**, not results calculated from the uploaded assessment.

False Negatives

A **False Negative (FN)** occurs when:

Actual patient = Diabetic
Model prediction = Non-Diabetic

False negatives are particularly dangerous in healthcare because a diabetic patient may not receive early diagnosis or treatment.

Reducing False Negatives

The model should prioritise **high recall/sensitivity**.

Possible methods:

1. Increase the class weight of diabetic cases.
2. Use SMOTE.
3. Adjust the classification threshold.
4. Perform hyperparameter tuning.
5. Use cost-sensitive learning.
6. Evaluate using Recall and F1-score rather than accuracy alone.

Clinically, reducing false negatives is important because missing a diabetic patient can delay treatment and increase the risk of complications.

2(c) Post-Pruning

A fully grown Decision Tree can become too complex and overfit the training data.

Post-pruning removes unnecessary branches after the tree has been constructed.

One suitable method is **Cost-Complexity Pruning**.

The objective is:

where:

- = classification error
- = number of terminal nodes

- = complexity parameter

A larger produces a simpler tree.

Example Comparison

Model	Accuracy	Complexity
Pre-pruned Tree	86%	Low
Post-pruned Tree	89%	Moderate
Fully grown Tree	93% training accuracy	High

The fully grown tree may have high training accuracy but poor generalisation.

Appropriate Model for Clinical Deployment

The **post-pruned Decision Tree** is more appropriate because:

- It reduces overfitting.
- It is easier to interpret.
- It provides simpler decision rules.
- It improves generalisation.
- Its decisions can be explained to healthcare professionals.

In medical applications, **interpretability and reliability** are extremely important.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Logistic Regression

Logistic Regression can be used as the statistical learning model.

It predicts the probability of diabetes:

where:

For this problem:

The output is a probability between 0 and 1.

For example:

$$P(\text{Diabetes}) = 0.82$$

If the threshold is 0.5:

$$0.82 > 0.5 \rightarrow \text{Diabetic}$$

Comparison with Decision Tree

Illustrative results:

Metric	Decision Tree	Logistic Regression
Accuracy	89%	91%
AUC-ROC	0.89	0.93

Again, these are **example values** because the assessment does not provide the actual patient dataset.

AUC-ROC

ROC represents the relationship between:

- True Positive Rate

- False Positive Rate

AUC measures the model's ability to distinguish diabetic and non-diabetic patients.

AUC closer to 1 indicates better discrimination.

When Logistic Regression is Preferable

Logistic Regression is preferable when:

- Interpretability is important.
- Relationships are approximately linear.
- Probability estimates are required.
- The dataset is not extremely complex.
- Healthcare professionals need understandable risk factors.

For example, the model can show how glucose and BMI contribute to diabetes probability.

3(b) Feature Importance Analysis

Feature importance can be determined separately for the Decision Tree and Logistic Regression.

Example

Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose
2	BMI	BMI
3	Age	Age

Therefore, both models may identify:

1. Glucose
2. BMI
3. Age

as the most important predictors.

Do They Agree?

In this illustrative example, **yes**.

The agreement indicates that these variables consistently contain strong predictive information.

However, the two models measure importance differently.

Decision Tree

Importance is based mainly on the reduction in impurity produced by a feature.

Logistic Regression

Importance can be interpreted through regression coefficients and odds ratios.

Therefore, even if the exact ranking differs, a feature that appears highly important in both models provides stronger evidence of its predictive relevance.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

The assessment specifically asks to model treatment recommendation as an **MDP** and apply Q-Learning for at least five patient episodes.

4(a) MDP Model

A **Markov Decision Process (MDP)** consists of:

where:

- = States
- = Actions
- = Transition probability
- = Reward
- = Discount factor

States

Patient health can be represented using states such as:

State	Description
S0	Healthy / Low Risk
S1	Moderate Risk
S2	High Risk
S3	Diabetes Controlled
S4	Diabetes Uncontrolled

Actions

The actions given in the assessment are:

- Diet
- Exercise
- Medication
- Monitor

Reward Function

A positive reward can be provided when the patient's health improves.

Example:

Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
No change	0
Health deterioration	-10

Outcome

Reward

Unsafe action

-20

Transition Dynamics

The transition represents how the patient's health changes after an action.

Example:

High Risk + Exercise → Moderate Risk

Moderate Risk + Diet → Controlled

High Risk + Medication → Controlled

Low Risk + Monitor → Low Risk

The actual transition probabilities should be learned from clinically valid longitudinal data rather than assumed in a real deployment.

4(b) Q-Learning

Q-Learning is a model-free reinforcement learning algorithm.

The Q-value update equation is:

where:

- = current Q-value
- = learning rate
- = reward
- = discount factor
- = next state

Assume:

Initially:

Episode 1 – Iteration 1

Suppose:

State = High Risk

Action = Exercise

Reward = +5

Next State = Moderate Risk

Suppose maximum future Q-value is 2.

Then:

Episode 2 – Iteration 1

Suppose:

State = Moderate Risk

Action = Diet

Reward = +10

Next State = Controlled

Maximum future Q-value = 3

Episode 2 – Iteration 2

Now suppose:

S1 → Diet → S3

Reward = +10

Current Q-value = 6.35

Assume:

Then:

Example Q-Table

After multiple episodes, an illustrative Q-table could be:

State	Diet	Exercise	Medication	Monitor
S0 Low Risk	.2 ⁴	5.1	1.2	6.8
S1 Moderate Risk	.9 ⁹ .98	7.4	5.8	2.3
S2 High Risk	.5 ⁵ .6	8.9	8.2	1.5
S3 Controlled	.4 ⁴ .5	6.2	3.1	7.1
S4 Uncontrolled	.2 ² .4	5.7	9.4	1.1

These values are illustrative.

Five Patient Episodes

A training sequence could be:

Episode	Initial State	Action Sequence	Final State
1	High Risk	Exercise → Diet	Controlled
2	Moderate Risk	Diet → Exercise	Controlled
3	High Risk	Medication → Monitor	Controlled
4	Uncontrolled	Medication → Diet	Moderate
5	Moderate Risk	Exercise → Diet	Controlled

The Q-table is updated after every state-action transition.

Convergence Criterion

Training can be considered converged when the change in Q-values becomes very small:

For example:

The algorithm can stop when the maximum Q-value change remains below 0.01 for several consecutive episodes.

4(c) Final Learned Policy

The policy is obtained by selecting the action with the highest Q-value:

Using the illustrative Q-table:

Patient State	Best Action
S0 Low Risk	Monitor
S1 Moderate Risk	Diet
S2 High Risk	Exercise
S3 Controlled	Monitor
S4 Uncontrolled	Medication

Thus:

Low Risk → Monitor

Moderate Risk → Diet

High Risk → Exercise

Controlled → Monitor

Uncontrolled → Medication

Important: In a real medical system, these actions must not be treated as autonomous treatment prescriptions. The policy would require clinical validation, physician oversight, safety constraints, and approved patient data.

Ethical Considerations

1. Patient Safety

A reinforcement-learning agent can make incorrect decisions if it learns from poor or incomplete data.

Therefore:

- Human clinical supervision is required.
- Unsafe actions must be prohibited.
- The system should include safety constraints.
- High-risk decisions should be reviewed by qualified healthcare professionals.

2. Bias

If the training dataset is biased, the AI may produce biased recommendations.

For example, if a particular demographic is underrepresented, the model may perform poorly for that group.

Therefore:

- Use diverse datasets.

- Evaluate performance across demographic groups.
- Monitor fairness continuously.

3. Explainability

Healthcare professionals need to understand why a particular recommendation was generated.

The system should provide:

- Patient risk factors
- Current health state
- Recommended action
- Reason for recommendation
- Confidence/risk information

4. Privacy

Patient medical information is highly sensitive. The system should use:

- Secure storage
- Encryption
- Access control
- Proper authentication
- Data anonymisation where appropriate

5. Human-in-the-Loop

The AI should act as a **decision-support system**, not an independent doctor.

The final treatment decision should remain with qualified healthcare professionals.

Overall Conclusion

The proposed system combines **inductive learning, Decision Trees, statistical learning and Reinforcement Learning** to create an intelligent diabetes-management framework. The Decision Tree can classify patients based on important medical features such as glucose, BMI and age. Logistic Regression provides interpretable probability-based risk prediction, while Q-Learning can model sequential treatment recommendations using patient health states, actions and rewards.